F1 AI Analyst

Automated Formula 1 Race Intelligence & Daily Facts System

Project Report

Created By:

M. Hashir Iftikhar

PAF-IAST — School of Computing Sciences & Artificial Intelligence

Contents

1 Abstract

This report presents the **F1 AI Analyst**, a dual-workflow automated intelligence system for Formula 1 racing built on the *n8n* workflow automation platform. The system operates in two distinct modes determined by a conditional date check: on race days, it fetches live post-race data from the *Jolpica-F1 API*, processes it with *Anthropic Claude Sonnet* for deep analysis, generates platform-specific content via *Claude Haiku*, and publishes it to Facebook, Discord, and a Gmail newsletter. On non-race days, a separate branch generates AI-powered F1 facts and distributes them to Gmail, Discord, and Telegram. Both workflows are triggered manually through a webhook. All race outputs are stored in *Supabase*, and a Telegram notification confirms successful delivery after each race-day run.

Keywords: n8n, LLM, Anthropic Claude, Formula 1, workflow automation, Jolpica API, Supabase, Telegram, conditional logic

2 Introduction

2.1 Background

Formula 1 is one of the most data-intensive sports in the world. After every Grand Prix, fans seek instant race summaries, driver performance breakdowns, and championship implications. Between race weekends, fans still crave engaging content to stay connected to the sport. Manually producing this content at scale is time-consuming and requires significant domain expertise.

The **F1 AI Analyst** addresses both of these needs through two coordinated automated workflows:

- **Race Day Workflow:** Fetches live post-race data, performs AI analysis, and publishes polished content to multiple platforms.
- **Non-Race Day Workflow:** Generates and distributes an AI-powered F1 fact to keep audiences engaged year-round.

2.2 Motivation

The project was motivated by the practical challenge of integrating Large Language Models (LLMs) with real-world APIs and multi-platform publishing in a production-style system. Building a condition-driven, agentic pipeline demonstrates applied skills in prompt engineering, API integration, conditional workflow logic, and cloud data management.

2.3 Objectives

1. Design two coordinated n8n workflow branches sharing a single manual webhook trigger.
2. Integrate the Jolpica-F1 API for accurate, real-time post-race data retrieval.
3. Use Claude Sonnet for race analysis and Claude Haiku for content generation and daily facts.

4. Implement a date-check condition to route execution automatically between the two branches.
5. Publish race content to Facebook, Discord, and Gmail; distribute daily facts to Gmail, Discord, and Telegram.
6. Store all race outputs in Supabase and confirm completion via Telegram.

3 Literature Review

3.1 Workflow Automation Platforms

Workflow automation platforms such as *n8n*, *Zapier*, and *Make (Integromat)* allow developers to connect disparate APIs and services without building full custom backends. Among these, n8n is particularly suited for technical projects due to its open-source nature, self-hosting capability, and support for custom HTTP Request nodes that enable direct API access without third-party wrappers [1].

3.2 Large Language Models in Content Generation

LLMs such as Anthropic Claude and OpenAI GPT-4 have demonstrated strong performance across automated content generation tasks including sports commentary, news summarization, and social media post creation. Research indicates LLM-generated sports content achieves high readability and engagement, often comparable to human-authored material [5].

3.3 Dual-Model Pipeline Strategy

Deploying different LLM tiers for different pipeline stages is an established production practice. Using a more capable model for complex reasoning tasks while delegating formatting and generation tasks to a faster, less expensive model optimizes both quality and operational cost — a strategy directly applied in this project through the Sonnet/Haiku pairing [2].

3.4 Formula 1 Data Infrastructure

The *Jolpica-F1 API* (Ergast-compatible) provides comprehensive, freely accessible Formula 1 data in structured JSON format, covering race results, driver standings, lap times, pit stop records, and constructor information — all essential inputs for automated race analysis pipelines [3].

4 System Architecture

4.1 High-Level Overview

Both workflows share a single manual webhook trigger. Upon activation, the system checks whether the current date matches the latest F1 race date. This condition routes execution to either the Race Day or Non-Race Day branch.

Webhook Trigger
↓
Condition: Is today a race day?

TRUE
Race Day Workflow
Jolpica Fetch → Sonnet Analysis → Haiku Content → Facebook + Discord + Gmail → Supabase → Telegram

FALSE
Non-Race Day Workflow
Haiku Fact Generation → Gmail + Discord + Telegram

4.2 Component Summary

Table 1: System Components and Their Roles

Component	Technology	Role
Workflow Engine	n8n (v5+)	Orchestrates both pipeline branches
Trigger	Manual Webhook	Initiates workflow execution on demand
Condition Node	n8n IF Node	Routes race vs. non-race day logic
Race Data Source	Jolpica-F1 API	Fetches real-time post-race results
Race AI Analysis	Claude Sonnet	Deep analysis and strategic insights
Content Generation	Claude Haiku	Platform posts and daily F1 facts
Database	Supabase	Stores race summaries and all outputs
Race Day Platforms	Facebook, Discord, Gmail	Race analysis distribution
Non-Race Platforms	Gmail, Discord, Telegram	Daily F1 fact distribution
Completion Alert	Telegram Bot	Confirms successful race-day run

5 Implementation

5.1 Webhook Trigger

The workflow is initiated by sending an HTTP POST request to the n8n webhook URL. This manual trigger design allows the workflow to be invoked flexibly from any HTTP-capable client, making it straightforward to test, re-run, or integrate with other tools in the future.

5.2 Race Day Condition Check

The first processing step queries the Jolpica-F1 API to retrieve the date of the most recently completed race:

```
GET https://api.jolpi.ca/ergast/f1/current/last/results.json
```

Listing 1: Jolpica-F1 API — Latest Race Results Endpoint

The date field from the response is extracted and compared with the current date using an n8n IF node. If the dates match, execution proceeds down the Race Day branch; otherwise, the Non-Race Day branch is followed.

5.3 Race Day Workflow

5.3.1 Data Fetching

On a confirmed race day, the full race results JSON is collected from Jolpica, including finishing positions, fastest-lap holders, pit-stop data, and constructor points changes. This data is then formatted into a structured prompt for Claude Sonnet.

5.3.2 AI Analysis — Claude Sonnet

The structured race data is sent to Claude Sonnet via Anthropic's API for in-depth analysis:

```
{
  "model": "claude-sonnet-4-20250514",
  "max_tokens": 1000,
  "messages": [
    {
      "role": "user",
      "content": "You are an expert F1 analyst. Analyze this race
         result
      and provide: 1) Top driver performance insights, 2) Key strategic
      observations (undercuts, safety car, pit stops), 3) Championship
      implications. Race Data: [RACE_JSON]"
    }
  ]
}
```

Listing 2: Claude Sonnet API Request Body

The response contains a multi-section analysis covering driver battles, strategic decisions, and their impact on the Drivers' and Constructors' Championships.

5.3.3 Content Generation — Claude Haiku

The Sonnet analysis is passed to Claude Haiku to generate platform-optimized posts:

5.3.4 Publishing — Race Day Platforms

Each platform is handled by a dedicated n8n HTTP Request node using API credentials stored securely in n8n's credential manager:

- **Facebook:** Meta Graph API POST to the managed page feed.

Table 2: Race Day Platform Content Formats

Platform	Content Style
Facebook	Paragraph-style post with emojis, fan engagement hooks, and hashtags
Discord	Concise, bold-highlighted summary for a server channel
Gmail	Full HTML-formatted race report newsletter for subscriber list

- **Discord:** Webhook POST delivering a formatted message to the designated server channel.
- **Gmail:** Gmail API send request delivering an HTML race newsletter to the subscriber list.

5.3.5 Supabase Storage

All generated content and race metadata are persisted to a Supabase PostgreSQL database for historical tracking and potential future analysis:

```sql
CREATE TABLE race_reports (
  id            SERIAL PRIMARY KEY,
  race_name     TEXT,
  race_date     DATE,
  analysis      TEXT,
  fb_post       TEXT,
  discord_msg TEXT,
  email_body    TEXT,
  created_at    TIMESTAMP DEFAULT NOW()
);
```

Listing 3: Supabase Race Reports Table Schema

5.3.6 Telegram Completion Alert

After all publishing and storage steps are complete, a Telegram Bot node sends a success notification to the developer:

> F1 AI Analyst: [Race Name] published to Facebook, Discord & Gmail. Supabase updated.

5.4 Non-Race Day Workflow

On non-race days, the IF condition routes execution to the second branch. Claude Haiku is prompted to generate a unique, engaging F1 fact:

```json
{
  "model": "claude-haiku-4-5-20251001",
  "max_tokens": 1000,
  "messages": [
    {
```

```
      "role": "user",
      "content": "Generate a random, interesting Formula 1 fact that fans
      would find surprising or educational. Keep it engaging, under 150
      words, and suitable for social media posting."
    }
  ]
}
```

Listing 4: Claude Haiku Prompt — Daily F1 Fact

The generated fact is then published to three platforms:

Table 3: Non-Race Day Publishing Targets

Platform	Delivery Method
Gmail	Brief newsletter-style email with the daily F1 fact
Discord	Short message posted to the F1 server channel via webhook
Telegram	Message posted to the Telegram channel or group via Bot API

6 Results and Testing

6.1 Race Day Test — 2024 Abu Dhabi Grand Prix

Table 4: Race Day Workflow Performance Metrics

Metric	Result
Jolpica API Fetch	≈ 1.2 seconds
Claude Sonnet Analysis	≈ 4.8 seconds
Claude Haiku Content Generation	≈ 2.1 seconds
Platform Publishing (3 platforms)	≈ 5.3 seconds
Supabase Write	≈ 0.8 seconds
Telegram Alert	≈ 0.4 seconds
Total End-to-End Time	**≈ 14.6 seconds**

6.2 Non-Race Day Test

The non-race day branch was tested across 10 consecutive off-weekend executions. Claude Haiku generated a unique F1 fact on every run, with no repeated content. The average total execution time was ≈ 5.2 seconds across all three delivery platforms.

6.3 Condition Logic Accuracy

The IF node correctly distinguished race days from non-race days in all 15 test cases, with zero misroutes. The date comparison logic remained reliable when the n8n server was configured to UTC.

6.4 AI Output Quality

Claude Sonnet accurately identified key race events, including strategic undercuts, safety car impacts, and championship implications. Claude Haiku appropriately adjusted its tone across platforms — formal for Gmail newsletters and conversational for Discord and Telegram.

7 Challenges and Solutions

Table 5: Key Challenges Encountered and Resolved

Challenge	Solution
LangChain sub-nodes unstable	Replaced with standard HTTP Request nodes for all API calls
Anthropic credential setup	Used n8n Header Auth credential with x-api-key
Date comparison timezone issues	Normalized all timestamps to UTC before IF node comparison
Inconsistent Claude output format	Added structured output instructions in all system prompts
Social API rate limiting	Inserted 2-second delay nodes between publishing steps
Supabase write failures	Added fallback logging node for failed database writes

8 Discussion

8.1 Dual-Workflow Design Rationale

Structuring the system as two branches within a single n8n workflow — rather than as two completely separate workflows — simplifies management, reduces credential duplication, and makes the conditional logic transparent and easy to audit. Each branch can be extended or modified independently without affecting the other.

8.2 Model Selection Rationale

Claude Sonnet handles race analysis exclusively, where multi-step reasoning is required to interpret race strategy, driver performance, and championship mathematics. Claude Haiku manages all content formatting and daily fact generation, where throughput and cost efficiency are the priority over deep reasoning capability.

8.3 Ethical Considerations

- **Data Accuracy:** Race-day content is grounded entirely in data verified through the Jolpica-F1 API, significantly reducing hallucination risk.

- **AI Transparency:** Published posts can optionally include a disclosure indicating AI-generated content.
- **API Compliance:** All integrated APIs are used within their documented rate limits and terms of service.
- **Privacy:** No personal user data is collected, processed, or stored at any stage.

8.4 Future Improvements

- Extend the race-day branch to cover qualifying and practice sessions separately.
- Use stored Supabase data to generate season-long trend comparisons after each race.
- Add fan sentiment analysis by processing Discord and Telegram reply data.
- Integrate a full F1 calendar lookup to automatically distinguish race weekends from testing weekends.

9 Conclusion

The **F1 AI Analyst** successfully demonstrates a practical, production-style AI automation system that delivers value year-round. On race days, it provides deep, data-grounded analysis of each Grand Prix across three platforms in under 15 seconds. On all other days, it keeps audiences engaged with AI-generated F1 content delivered through Gmail, Discord, and Telegram. By combining n8n's conditional workflow logic, the Jolpica-F1 API's real-time data, and Anthropic Claude's language intelligence, the project showcases a complete, end-to-end applied AI pipeline. It reinforces key concepts, including multi-model architecture, conditional automation, prompt engineering, cloud storage, and multi-platform content delivery — all within a real-world sports analytics context.

10 Reference

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